

Battery Balance & Management Board: Firmware Specifications

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Board: PSoC

Firmware Engineers: Prerana

Firmware Project Overview

The Battery Balance & Management board is used on the rover to balance the two batteries' power output. On the firmware side, the responsibility will be primarily monitoring the health of the batteries by gathering current data using the INA226. Currently, there is an [existing driver for the INA226](#) current sensor, though it may not be complete, so this board's firmware team, along with the power distribution board firmware teams, will be responsible for debugging and validating the driver.

Background Information

- [Battery_Management_Board_PY2024](#): the previous year's battery management firmware, which I wrote last year; though it was pretty messy and unstructured, so it might not be the best idea to base the new firmware on this code
- [INA226 Driver](#): driver for interfacing with the INA226 current sensor
- [HindsightCAN Wiki](#): details the CAN packets used and format; most likely this board will use "Telemetry Pull" packets for current
- [Battery Balance & Management Hardware Spec \(PY2024\)](#)

Expected Behavior

Primary Goals

- ☐ Upon receiving a "Telemetry Pull" CAN packet for current, the current data from the INA226 sensor will be sent to the requester via CAN packet
 - ☐ The format of the data needs to be discussed with the software team
 - ☐ If there is an error retrieving current data or interfacing with an INA226, then print an error to the debug UART and send a "Failure Report" CAN packet
- ☐ If a "Telemetry Pull" CAN packet is received but requests data that is not the current or chip type, then print an error to the debug UART and send a "Failure Report" CAN packet

Secondary Goals

- ☐ If the current is low, then respond by lighting status LEDs if available or sending a CAN packet
 - ☐ Need to discuss with hardware what the low current threshold would be
- ☐ Standard CAN LED behavior
- ☐ Standard DBG LED behavior

Minimum Firmware Test Cases

- ☐ “Telemetry Pull” CAN packet for current
- ☐ “Telemetry Pull” CAN packet for unsupported data
- ☐ “Telemetry Pull” CAN packet with unexpected INA226 behavior (OPTIONAL: may be difficult to test)

Recommended Initial Steps

- The [Battery Balance & Management Hardware Spec](#) from last year can explain what the board does on a hardware level
- Look at the [INA226 Driver](#) to see what has been completed or what needs to be validated
 - As far as I’m aware, we can get voltage readings, but not current